Risk Assessment & Mitigation Plan

1. Purpose

This document identifies the main risks that could affect the delivery of portfolioGenie (an AI-powered portfolio generator), scores them, and defines how the team will prevent, reduce or respond to each one. It supports the Gantt chart in Project_Plan.xlsx and the DEPI deadlines (Planning 16 Oct, Design 6 Nov, Implementation 30 Nov, Final delivery 4 Dec 2026).

2. Scoring Method

Each risk is rated for Probability (1 = very unlikely, 5 = almost certain) and Impact (1 = negligible, 5 = project failure). Risk Score = Probability x Impact.

Score	Level	Required action
15 - 25	**Critical**	Act immediately; PM reviews weekly
10 - 14	**High**	Mitigation plan active; reviewed weekly
5 - 9	**Medium**	Monitor; reviewed every two weeks
1 - 4	**Low**	Accept; review at phase end

3. Risk Summary

Total risks identified: 14 | Critical: 1 | High: 6 | Medium: 7 | Low: 0

4. Risk Register

ID	Category	Risk description	Prob.	Impact	Score	Level	Mitigation (prevent / reduce)	Contingency (if it happens)	Owner
R1	Schedule	Tight DEPI deadlines; tasks overrun (esp. Implementation by 30 Nov, with Frontend, Backend and AI built one after another)	4	4	16	**Critical**	Weekly progress check against Gantt; MoSCoW prioritization; 1-week buffer before each DEPI deadline	Cut Could-have features; add parallel work	Rashed / Hazem
R2	Scope	Scope creep - extra portfolio features added mid-project	4	3	12	**High**	Freeze requirements after SRS (16 Oct); change-request log approved by PM	Move extras to a 'future work' list	Rashed / Walaa
R3	Technical	LLM output quality is inconsistent or inaccurate for generated portfolio content	3	4	12	**High**	Structured prompts + templates; human edit step; test with sample profiles; start prompt prototyping during Design phase because the AI build window is short (24-27 Nov)	Fallback to template-based generation with user-filled fields	AI
R4	Technical	AI API cost, rate limits or downtime	3	4	12	**High**	Use free/low-tier quota; cache results; retry & timeout handling	Switch provider or use a smaller/open model	AI / Backend (TBD)
R5	Technical	Integration problems: frontend is built first on mock data, so it may not match the real backend/AI responses	3	3	9	**Medium**	Agree API contract (JSON format) before frontend starts; mock data follows it; weekly integration builds once backend starts	Dedicated integration window (28-30 Nov); keep mock-data version as fallback for the demo	Backend
R6	Resource	Team member unavailable (illness, exams, other commitments)	3	4	12	**High**	Cross-train on key modules; document work; shared repo	Reassign tasks; reduce scope of non-core features	Rashed
R7	Resource	Skill gaps with chosen technologies / AI tools	3	3	9	**Medium**	Early spikes during Literature Review; tutorials; mentor support	Choose simpler stack	All
R8	Requirements	Requirements unclear or change after user feedback	3	3	9	**Medium**	Survey real users; validate SRS with mentor; prototype early	Re-prioritize backlog in change meeting	Walaa / Mostafa
R9	Quality	Insufficient testing time before 4 Dec	3	4	12	**High**	Start test cases in design phase; test continuously during implementation	Focus testing on critical user flows	QA
R10	Security / Legal	Leakage of user personal data; AI-generated text copyright concerns	2	5	10	**High**	Minimal data collection; HTTPS; env secrets; privacy notice; check content licences	Delete data on request; remove disputed content	Backend
R11	Infrastructure	Hosting/deployment failure or data loss before demo	2	4	8	**Medium**	Git + regular backups; deploy early (staging); documented deployment steps	Run demo locally / recorded demo as backup	Backend
R12	Communication	Miscommunication or delayed decisions inside team and with DEPI mentor	3	2	6	**Medium**	Weekly meeting, shared task board, single source of truth for documents	Escalate to mentor; daily sync in critical weeks	Rashed
R13	Technical	Website looks or works differently across browsers (Chrome, Edge, Safari) and on mobile screens	3	3	9	**Medium**	Responsive design from the start (mobile-first); test on 3 browsers and 2 screen sizes each sprint	Fix critical layout bugs first; drop non-essential animations/features	Frontend (TBD)

R14	Performance	Website loads slowly or AI generation makes pages hang (heavy images, long API calls)	3	3	9	**Medium**	Optimize images, lazy loading, loading indicators, API timeouts; measure load time in testing	Cache generated results; simplify templates	Frontend / Backend (TBD)

5. Monitoring & Review

- The PM reviews the risk register in the weekly team meeting and updates probability, impact and status.
- Any risk that reaches Critical triggers an immediate meeting and is reported to the DEPI mentor.
- New risks are added with an ID, owner and mitigation; closed risks are kept in the log with the closing date.
- Risk review checkpoints: end of each phase (16 Oct, 6 Nov, 30 Nov, 4 Dec 2026).

6. Resource & Schedule Coordination

Owners for planning, research and requirements risks are named team members (Hazem, Hind, Walaa, Mostafa, Rashed); owners shown as Backend, AI, Frontend or QA are roles to be assigned once the implementation split is agreed (see the Resource Plan sheet in Project_Plan.xlsx). A one-week buffer is kept before each DEPI deadline, and the last week of November is reserved for integration.